Question 36 Solution

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Question

Given the Boolean expression:

$[X + Z(Y + (Z + XY)')][(\overline{Y} + \overline{Z}(X + Y))]$

If $X = 1$, simplify the expression.

Solution

Substitute $X = 1$

$[1 + Z(Y + (Z + Y)')][(\overline{Y} + \overline{Z}(1 + Y))]$

Since $1 + A = 1$,

First bracket simplifies to 1.

Second bracket simplifies to $(\overline{Y} + \overline{Z})$.

Final simplified expression:

$F = \overline{YZ}$

Thus, the circuit behaves as a NAND gate.

Truth Table

Y	Z	$F = \overline{YZ}$
0	0	1
0	1	1
1	0	1
1	1	0

Conclusion

The output is HIGH except when both Y and Z are HIGH. Therefore, the simplified logic is NAND.